

Aspose.Pdf Native Windows Viewer Demo & Test Document

A native high-performance PDF desktop viewer built with WPF and .NET 9

This sample document demonstrates the viewing capabilities of the application:

- Continuous Vertical Scrolling & Single Page Paginated Modes
- Dynamic Zooming (Fit Width, Fit Page, 25% - 500%) & Mouse Wheel Zoom
- In-Document Text Search with Real-Time Match Counter
- Hierarchical Bookmark / Outline Navigation
- Page Thumbnails with Active Page Indicator
- Document Properties & Metadata Inspector
- Export Pages to High-Resolution PNG/JPEG Images
- High-Quality Windows Printing
- Modern Light & Dark Themes

1. Architecture & Capabilities

The desktop viewer renders PDF documents directly via Aspose.Pdf rasterization devices. Rendering is offloaded onto background thread pools with an LRU (Least-Recently-Used) cache, guaranteeing responsive 60 FPS scrolling and low memory consumption even when navigating hundreds of pages.

Search functionality utilizes TextFragmentAbsorber to pinpoint exact occurrences of text across the entire document structure, highlighting matches with page jumps.

2. Technical Specifications

Component	Specification
Target Framework	.NET 9.0 Windows (net9.0-windows)
UI Framework	Windows Presentation Foundation (WPF)
Solution Format	Modern .slnx (Visual Studio 2022+ / .NET 9+)
PDF Engine	Aspose.Pdf for .NET
License System	Aspose.Total.lic Automatic Discovery & Embedded Resource
Cache Mechanism	Thread-safe LRU Memory Bitmap Cache
Theme Support	Light & Dark Mode dynamically switched

3. Caching & Background Threading

The rendering engine combines WPF dispatcher prioritization with background task workers. Rendered bitmaps are frozen (`Freezable.Freeze()`) to permit safe cross-thread sharing without UI deadlocks.

An LRU memory cache ensures that navigating large documents with dozens or hundreds of pages operates with a constant memory footprint, automatically recycling bitmaps when memory capacity is reached.

4. Outline Hierarchy & Bookmarks

Document outlines provide direct hierarchical navigation through chapters, sections, and appendices. Selecting any item in the left sidebar instantly centers the viewport on the destination page and updates the active indicator.

5. High-Speed Text Search

Keyword search across all pages executes asynchronously, returning snippets, page numbers, and hit counters. Users can double-click any search result to jump directly to that page.

Keyword for search verification: QuantumComputingX99

6. Image Export & Printing

Export pages to standalone PNG or JPEG files at customizable DPI resolutions (72 to 600 DPI). Standard Windows Print Dialog integration allows printing full documents or custom page ranges with native quality.

7. Summary & Verification Notes

All core features have been automated and tested:

- Unit tests passing with 100% success rate.
- Aspose.Total license active without watermarks or page limits.
- Smooth continuous scrolling across all pages.
- Navigation sidebar with real-time center page synchronization and thumbnail scrolling.